

- Question 1:** (a) $[18] = \{ 4n + 2 \mid n \in \mathbb{Z} \}$.
 (b) $[31] = \{ 4n + 3 \mid n \in \mathbb{Z} \}$.
 (c) There are 4 equivalence classes.

Question 2: *Proof.* We prove that $\equiv_n$ on $\mathbb{Z}$ is an equivalence relation by showing that it is reflexive, transitive, and symmetric. Suppose $n \in \mathbb{Z}^+$.

Reflexive Let a be an arbitrary element in $\mathbb{Z}$. Clearly, $a - a = 0 = kn, k \in \mathbb{Z}$. So, $\forall a \in \mathbb{Z}, a \equiv_n a$.

Symmetric Let a, b be arbitrary elements of $\mathbb{Z}$. Suppose $a \equiv_n b$. By definition, $a - b = kn, k \in \mathbb{Z}$. Multiplying the equation by -1 , we get

$$b - a = -kn.$$

$-k$ is still an integer, and thus by definition, $b \equiv_n a$.

Transitive Let a, b, c be arbitrary elements of $\mathbb{Z}$. Suppose $a \equiv_n b$ and $b \equiv_n c$. By definition, we have $a - b = kn, k \in \mathbb{Z}$ and $b - c = gn, g \in \mathbb{Z}$. Then, $b = a - nk$. Subbing in, we get that

$$\begin{aligned} a - kn - c &= gn \\ a - c &= (k + g)n \end{aligned}$$

By definition, we have $a \equiv_n c$.

Thus, we are done. □

Question 3: *Proof.* Suppose $a, b, c, d \in \mathbb{Z}$, and $a \equiv_n c$ and $b \equiv_n d$. Then,

$$\begin{aligned} a - c &= nk, k \in \mathbb{Z} \\ b - d &= nl, l \in \mathbb{Z}. \end{aligned}$$

Adding the equations, we get

$$a + b - c - d = n(k + l).$$

Simplifying, we get

$$(a + b) - (c + d) = n(k + l).$$

Thus, by definition $a + b \equiv_n c + d$. □

Question 4: (a) *Proof.* The theorem given in the proposition is equivalent to claiming that ‘If ab is even, then a is even or b is even’. We have proven this theorem to be true, so it is the case that if $ab \equiv_2 0$, then $a \equiv_2 0$ or $b \equiv_2 0$. □

- (b) The answer is no. $10 \times 2 = 20 \equiv_4 0$, but $10 \not\equiv_4 0$ and $2 \not\equiv_4 0$.
 (c) The answer to the question will be yes when n is a prime.

Proof. Suppose n is prime. Let $a, b \in \mathbb{Z}$. We proceed via the contrapositive, which is the statement

$$\text{if } a \not\equiv_n 0 \text{ and } b \not\equiv_n 0, \text{ then } ab \not\equiv_n 0.$$

Then, n is not a prime factor of either a or b . □

Question 5: *Proof.* We show three requirements.

- 1 $e \in A$ is its own inverse ($e * e = e$), and so it has an inverse, and is in $A^\times$.
- 2 Suppose $a \in A^\times$. Then, $a \in A$ and a has an inverse $a^{-1} \in A$. Similarly, a^{-1} has an inverse, namely a . So, $a^{-1} \in A$ and a^{-1} is invertible, so $a^{-1} \in A^\times$.
- 3 Suppose $a, b \in A^\times$. Then, $a, b \in A$, and $a * b \in A$. Let $c = (b^{-1} * a^{-1})$. Then, we compute

$$(a * b) * c = a * b * b^{-1} * a^{-1} = a * a^{-1} = e.$$

Therefore, $a * b$ has an inverse. Therefore, $a * b \in A^\times$. □

Question 6: *Proof.* We first show that matrix multiplication $(*)$ is a unital binary operation on $\text{Mat}_{2 \times 2}(\mathbb{R})$.

- 1 We are given that $*$ is associative.
- 2 Let $E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{Mat}_{2 \times 2}(\mathbb{R})$. We compute that

$$AE = EA = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Thus, $*$ is unital.

Next, we show that the set $\text{GL}_2(\mathbb{R})$ is actually the set of all the matrices in $\text{Mat}_{2 \times 2}(\mathbb{R})$ that are invertible. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We then reduce A to E augmented with E :

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right]$$

Let the resultant augmented matrix be A' . Then, we compute

$$AA' = E \text{ and } A'A = E.$$

A' exists only when $ad - bc \neq 0$, which is true for all $M \in \text{GL}_2(\mathbb{R})$. In other words, $\text{GL}_2(\mathbb{R}) = \{ A \in \text{Mat}_{2 \times 2}(\mathbb{R}) \mid A \text{ is invertible} \}$.

Lemma 1. *See the result of problem 5.*

Thus, $\text{GL}_2(\mathbb{R})$ is a group with respect to matrix multiplication. $\square$

Question 7:

(a) *Proof.* We show that R is reflexive, symmetric, and transitive.

Reflexive $*$ is unital, so there is an identity element e . $e * e = e$, so e is invertible and is its own inverse. So, $a = e * a * e$. Therefore, R is reflexive.

Symmetric Let $a, b \in A$ and aRb . Then, we compute

$$\begin{aligned} b &= t^{-1} * a * t \\ t^{-1} * b &= a * t \\ t^{-1} * b * t &= a. \end{aligned}$$

Let $t^{-1} = t'$. $t' \in A$ and is invertible. Thus, bRa , and R is symmetric.

Transitive Let $a, b, c \in A$. Suppose aRb and bRc . Then, $b = t^{-1} * a * t$ and $c = u^{-1} * b * u$ such that $t, u \in A$. Plugging in b for a , we get that

$$c = u^{-1} * t^{-1} * a * t * u.$$

Let $v = t * u$. We compute that

$$(t * u) * (u^{-1} * t^{-1}) = t * t^{-1} = e.$$

Thus, $u^{-1} * t^{-1} = v^{-1}$, and so $c = v^{-1} * a * v$, $v \in A$. Thus, R is transitive.

So, R is an equivalence relation. $\square$

(b) If $*$ is commutative, then the relation $aRb := b = t^{-1} * a * t$, $a, b, t \in A$ can be simplified to $b = a$. This means that two elements are related only if they are the same element. Thus, each element is its own equivalence class. Formally, the equivalence classes are given by $\{ [a] = a \mid a \in A \}$.

(c) We compute that the equivalence classes of D_6 are the following:

$$[e] = \{ e \}, [r] = \{ r, r^2 \}, [s] = \{ s, sr, sr^2 \}.$$